

SI Session 3 Review

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1 Solving First Order by Grouping

1.1 Problem 1

$$(2xy^2 - 3x)dy + (2y^3 + 3y)dx = 0$$

1.1.1 Solution

$$(2xy^2 dy + 2y^3 dx) = (3xdy - 3ydx)$$

$$y^2(2xdy + 2ydx) = (3xdy - 3ydx)$$

$$d(xy) = -d\left(\frac{3x}{y}\right)$$

$$\int d(xy) = \int -d\left(\frac{3x}{y}\right)$$

$$2xy = -\frac{3x}{y} + K_1$$

1.2 Problem 2

$$(14xy^3 dx + 21x^2 y^2 dy) = (8x^3 y^3 dx + 6x^4 y^2 dy)$$

1.2.1 Solution

$$7d(x^2 y^3) = 2d(x^4 y^3)$$

$$\int 7d(x^2 y^3) = \int 2d(x^4 y^3)$$

$$7x^2 y^3 = 2x^4 y^3 + K_1$$

2 Newton's Cooling

2.1 Problem 3

A body, whose temperature is 70°C at $t = 0$, cools at a rate (in $^{\circ}\text{C}$ /minute) that is equal to one-fourth the difference between the temperature of the body and the temperature of the surrounding air. If the temperature of the surrounding air remains constant at 30°C , (a) what is the temperature of the body at the end of 5 minutes, and (b) when does the temperature of the body reach 31°C ?

2.2 Solution

$$\frac{dT}{dt} = -\frac{1}{4}(T - 30)$$

$$\frac{dT}{dt} + \frac{1}{4}T = \frac{30}{4}$$

$$IF = e^{\int \frac{1}{4} dt} = e^{\frac{t}{4}}$$

$$T = 30 + K_1 e^{-\frac{t}{4}}$$

$$K_1 = 70 - 30 = 40$$

Part (a)

$$T = 30 + 40e^{-\frac{5}{4}}$$

Part (b)

$$31 = 30 + 40e^{-\frac{t}{4}}$$

3 First Order Bernoulli

3.1 Problem 4

$$\frac{dy}{dx} + \frac{2x}{x^2 + 1}y = \frac{x}{x^2 + 1}y^2$$

3.1.1 Solution

$$z = y^{1-n} = y^{-1}$$

$$\frac{dy}{dx} = -y^2 \frac{dz}{dx}$$

Make substitution

$$\frac{dz}{dx} - \frac{2x}{x^2 + 1}z = \frac{-x}{x^2 + 1}$$

$$IF = e^{\int \frac{-2x}{x^2+1} dx} = (x^2 + 1)^{-1}$$

$$\int d((x^2 + 1)^{-1}z) = \int \frac{-x}{(x^2 + 1)^2} dx$$

$$(x^2 + 1)^{-1}z = \frac{1}{2(x^2 + 1)} + K_1$$

$$y = \left(\frac{1 + K_2(x^2 + 1)}{2}\right)^{-1}$$

3.2 Problem 5

$$\frac{dy}{dx} + \frac{\sin(x)}{1 + \cos(x)} = \frac{e^x}{1 + \cos(x)} y^3$$

3.2.1 Solution

$$z = y^{-2}, \quad \frac{dy}{dx} = \frac{-y^3}{2} \frac{dz}{dx}$$

Make substitution

$$\frac{dz}{dx} - \frac{2\sin(x)}{1 + \cos(x)} z = \frac{-2e^x}{1 + \cos(x)}$$

$$IF = (1 + \cos(x))^2$$

$$\int d((1 + \cos(x))^2 z) = \int -2e^x(1 + \cos(x)) dx$$

Split the integral

$$\int d((1 + \cos(x))^2 z) = (1 + \cos(x))^2 z$$

$$\int -2e^x(1 + \cos(x)) dx = -2\left(\int e^x dx + \int e^x \cos(x) dx\right)$$

Use integration by parts on second integral

$$-2\left(\int e^x dx + \int e^x \cos(x) dx\right) = -2e^x - e^x \cos(x) - e^x \sin(x)$$

$$(1 + \cos(x))^2 z = -2e^x - e^x \cos(x) - e^x \sin(x)$$

Get rid of z and do some algebra to solve for y

$$y = \pm \sqrt{\frac{(1 + \cos(x))^2}{-2e^x - e^x \cos(x) - e^x \sin(x) + K_1}}$$

4 Growth and Decay

4.1 Problem 6

Under natural circumstances, the population of mice on a certain island increases at a rate that is proportional to the number of mice that are on the island at the end of t years. Between the end of 2000 and the end of 2005, there were no cats on this island, and during this 5-year time period the mice population doubled, reaching an all-time high of 300,000 mice at the end of this 5-year time period. Because of the rapidly increasing number of mice on the island, the islanders decide to import some cats to the island at the end of 2005 for the purpose of reducing the number of mice on the island. Assuming that the natural rate of birth of mice on the island is offset by the work of the cats after the end of 2005, with the cats killing 5000 mice per month thereafter, and also letting $t = 0$ correspond to the end of 2000, how many mice are on the island at the end of 2010?

4.2 Solution

$$\frac{dx}{dt} = kx$$

Solve by separation of variables

$$x = e^{kt}K_2$$

Use the ICs:

$$t = 0, x = 150,000$$

$$K_2 = 150,000$$

Use the ICs:

$$t = 5, x = 300,000$$

$$k = \frac{\ln(2)}{5}$$

Now solve when the cats enter the island

$$\frac{dx}{dt} = kx - 60,000$$

This is a linear DE

$$\begin{aligned} IF &= e^{\int -k dt} = e^{-kt} \\ \int d(e^{-kt}x) &= \int -60,000e^{-kt} dt \\ e^{-kt}x &= \frac{60,000e^{-tk}}{k} + K_3 \end{aligned}$$

Using the ICs:

$$t = 5, x = 300,000$$

$$K_3 = 3000,000e^{-5k} + \frac{60,000e^{-5k}}{k}$$

Given we know k , K_3 , and $t = 10$, the answer below is our final answer.

$$x = \frac{60000}{k} + \left(300000 - \frac{60000}{k} \right) e^{5k}$$